* Little omega :-

As we know big omega was giving the lower bound. This little omega will ~~also~~ also going to give lower bound. But it will not be tight. So, it's gonna be ~~like~~ loosely lower bound.

Eg :-

$$f = \cancel{\omega(g)}$$

$$f \lim_{N \to \infty} \frac{N^3}{N^2} = \lim_{N \to \infty} N = \infty$$

Big Ω

$f = \Omega(g)$

this means

$f \geq g$

lower ~~bong~~ bound.

{ It is giving a lower bound, It can increase more than that }

little ω

$f = \omega(g)$

this means

$f > g$ (Strictly greater) difference.

Read this at the last

Q: What if an algo has lb & up as (N^2).

$= O(N^2)$ & $\Omega(N^2)$

Theta Notation: (Combining both)

$\Theta(N^2) \Rightarrow$ words: Both up & lb is $= N^2$

$$\boxed{0 < \lim_{n \to \infty} \frac{f(n)}{g(n)} < \infty}$$

Maths:

$$\lim_{N \to \infty} \frac{f(N)}{g(N)} = \infty$$

Q When do use this?

Ans

When you want somthink like strictly greater or strictly less than etc.

Note: In reality we use Big-Oh notations only.